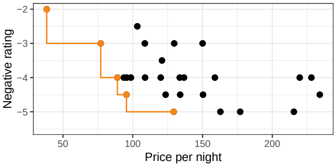
Optimization in Machine Learning

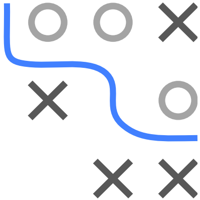
Multi-criteria Optimization
Introduction



Learning goals

- What is multi-criteria optimization?
- Motivating examples
- Pareto optimality

FINDING PARETO-OPTIMAL SOLUTIONS I



- **Weighted sum scalarization** solves:

$$\boldsymbol{x}^* := \arg\min_{\boldsymbol{x} \in \mathcal{S}} \sum_{i=1}^{m} w_i f_i(\boldsymbol{x})$$

We can show that this approach can find Pareto-optimal solutions.

- Without loss of generality we can assume $\sum_{i=1}^{d} w_i = 1$.
- **Theorem:** If $w_i > 0$ for all i, then $\boldsymbol{x}^*$ is Pareto-optimal. **Proof.**

Suppose $\boldsymbol{x}^*$ is not Pareto-optimal, i.e., there exists $\boldsymbol{x}$ such that $f_i(\boldsymbol{x}) \leq f_i(\boldsymbol{x}^*)$ for all i and $f_j(\boldsymbol{x}) < f_j(\boldsymbol{x}^*)$ for at least one j. Then,

$$\sum_{i=1}^{m} w_i f_i(\boldsymbol{x}) < \sum_{i=1}^{d} w_i f_i(\boldsymbol{x}^*),$$

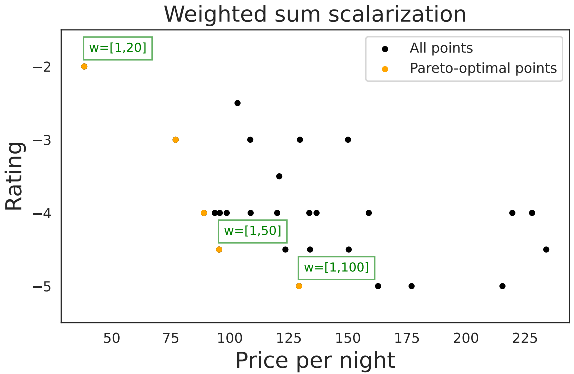
which contradicts the definition of $\boldsymbol{x}^*$ as the minimizer.

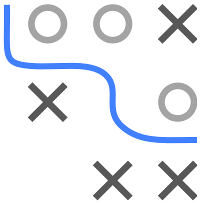
FINDING PARETO-OPTIMAL SOLUTIONS II

- Not all Pareto-optimal solutions can be found by solving

$$\boldsymbol{x}^* := \arg\min_{\boldsymbol{x} \in \mathcal{S}} \sum_{i=1}^{m} w_i f_i(\boldsymbol{x})$$

- **Example:** We find three out of the five Pareto-optimal solutions:



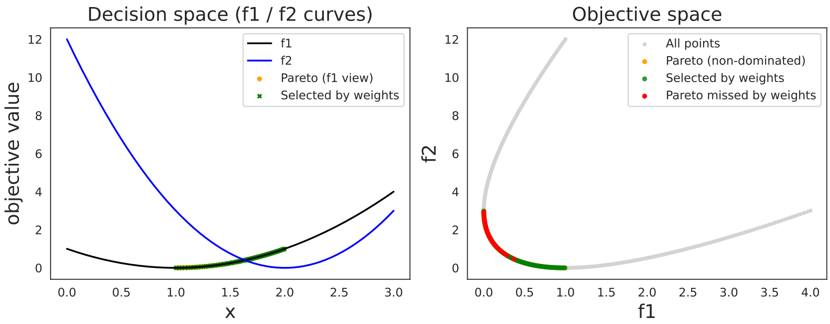


FINDING PARETO-OPTIMAL SOLUTIONS III

- **Example:** $f_1(x) = (x-1)^2$ and $f_2(x) = 3(x-2)^2$ with

$$\min_x f(x) = \left(f_1(x),\ f_2(x)\right), \quad 0 \leq x \leq 3.$$

- Use $0 \leq x \leq 3$ with 300 steps (100 non-dominated points).
- Use $w_1 \in (0, 1)$ with 100 steps and $w_2 = 1 - w_1$.



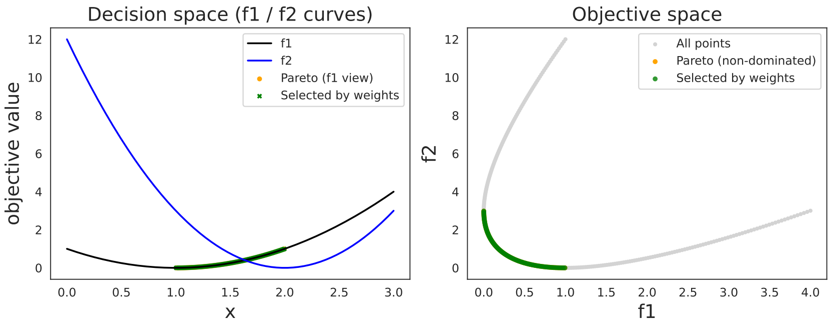
- Not all Pareto-optimal solutions are found.

FINDING PARETO-OPTIMAL SOLUTIONS IV

- **Example:** $f_1(x) = (x-1)^2$ and $f_2(x) = 3(x-2)^2$ with

$$\min_x f(x) = \big(f_1(x),\ f_2(x)\big), \quad 0 \le x \le 3.$$

- Use $0 \le x \le 3$ with 300 steps (100 non-dominated points).
- Using w_1 with 300 steps yields all Pareto-optimal solutions:



- Under which conditions can we find all Pareto-optimal solutions?

FINDING PARETO-OPTIMAL SOLUTIONS V

Theorem (Geoffrion, 1968): Let $\mathcal{S}$ be convex and assume $f_1, \dots, f_d$ are convex. Then $\boldsymbol{x}^* \in \mathcal{S}$ is properly Pareto-optimal if and only if $\boldsymbol{x}^*$ is an optimal solution of

$$\min_{\boldsymbol{x} \in \mathcal{S}} \sum_{i=1}^{m} w_i f_i(\boldsymbol{x})$$

with strictly positive weights w_i for all i.

- For convex functions and convex search spaces, we can find all properly Pareto-optimal solutions using weighted sum scalarization.
- **Proper Pareto-optimal solution**: Exclude solutions where an arbitrary small improvement in one objective leads to an arbitrary large deterioration in another objective.

FINDING PARETO-OPTIMAL SOLUTIONS VI

Definition (Geoffrion, 1968): A Pareto-optimal solution $\boldsymbol{x}^* \in \mathcal{S}$ is called **properly** Pareto-optimal, if there exists a constant $M > 0$ such that for all i and $\boldsymbol{x} \in \mathcal{S}$ with $f_i(\boldsymbol{x}) < f_i(\boldsymbol{x}^*)$ there exists an index j, such that $f_j(\boldsymbol{x}^*) < f_j(\boldsymbol{x})$ and

$$\frac{f_i(\boldsymbol{x}^*) - f_i(\boldsymbol{x})}{f_j(\boldsymbol{x}) - f_j(\boldsymbol{x}^*)} \leq M$$

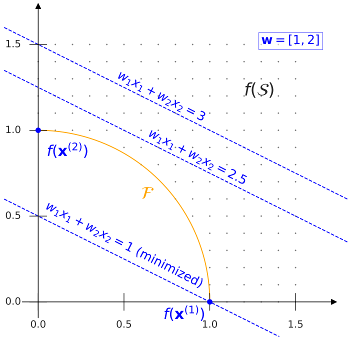
Example: $f_1(x) = (x-1)^2$ and $f_2(x) = 3(x-2)^2$ for $0 \leq x \leq 3$.

- Pareto-optimal solutions are $x \in [1, 2]$ (as exercise).
- $x^* = 1$ is not properly Pareto-optimal.
  Proof. For $x^* = 1$ we have $f_1(x^*) = 0$ and $f_2(x^*) = 3$. Let $x = 1 + \epsilon$ for $\epsilon > 0$, then $f_2(x) < f_2(x^*)$ and $f_1(x) > f_1(x^*)$. Then,

$$\frac{f_2(x^*) - f_2(x)}{f_1(x) - f_1(x^*)} = \frac{3 - 3(\epsilon - 1)^2}{\epsilon^2} = \frac{6\epsilon - 3\epsilon^2}{\epsilon^2} \to \infty \text{ for } \epsilon \to 0$$

FINDING PARETO-OPTIMAL SOLUTIONS VII

- Convexity is required to find properly Pareto-optimal solutions
- In non-convex settings: Often poor coverage of the Pareto front.
- **Example:** $\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \geq 1\}$ with $f_1(\boldsymbol{x}) = x_1$ and $f_2(\boldsymbol{x}) = x_2$.
- Pareto front: $\mathcal{F} = \{\boldsymbol{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$
- Weighted sum scalarization only finds $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$

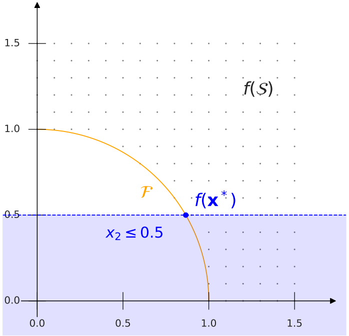


FINDING PARETO-OPTIMAL SOLUTIONS VIII

- **Alternative:** ϵ-constraint method

$$\boldsymbol{x}^* := \arg\min_{\boldsymbol{x}\in\mathcal{S}} f_1(\boldsymbol{x}) \quad \text{s.t. } f_2(\boldsymbol{x}) \leq \epsilon_2.$$

- By varying ϵ_i we can find different Pareto-optimal solutions:



FINDING PARETO-OPTIMAL SOLUTIONS IX

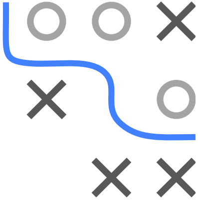
- General form of the ϵ-constraint method:

$$\boldsymbol{x}^* := \arg\min_{\boldsymbol{x}\in\mathcal{S}} f_i(\boldsymbol{x}) \quad \text{s.t. } f_j(\boldsymbol{x}) \leq \epsilon_j, \quad j = 1, \dots, m, j \neq i.$$

- Are solutions found by the ϵ-constraint method Pareto-optimal?
- **Theorem:** If $\boldsymbol{x}^*$ is the unique optimal solution of the ϵ-constraint method for some index i, then $\boldsymbol{x}^*$ is Pareto-optimal.

  Proof. Suppose $\boldsymbol{x}^*$ is not Pareto-optimal. Then there exists $\boldsymbol{x} \in \mathcal{S}$, such that $f_i(\boldsymbol{x}) \leq f_i(\boldsymbol{x}^*)$ for all $i = 1, \dots, m$ and $f_j(\boldsymbol{x}) < f_j(\boldsymbol{x}^*)$ for at least one $j \in \{1, \dots, m\}$. Since $\boldsymbol{x}^*$ is feasible for the ϵ-constraint problem, it follows that $f_i(\boldsymbol{x}) \leq f_i(\boldsymbol{x}^*) \leq \epsilon_i$, so $\boldsymbol{x}$ is also feasible. Moreover, $f_1(\boldsymbol{x}) \leq f_1(\boldsymbol{x}^*)$, and since $\boldsymbol{x}^*$ is the unique minimizer of cost_1, we must have $\boldsymbol{x} = \boldsymbol{x}^*$, which is a contradiction.

FINDING PARETO-OPTIMAL SOLUTIONS X



- Can the ϵ-constraint method find all Pareto-optimal solutions?

$$\boldsymbol{x}^* := \arg\min_{\boldsymbol{x}\in\mathcal{S}} f_i(\boldsymbol{x}) \quad \text{s.t. } f_j(\boldsymbol{x}) \leq \epsilon_j, \quad j = 1, \ldots, m, j \neq i.$$

- **Theorem:** $\boldsymbol{x}^* \in \mathcal{S}$ is Pareto-optimal if and only if there exists $\epsilon \in \mathbb{R}^m$ such that $\boldsymbol{x}^*$ is an optimal solution for all $i = 1, \ldots, m$.

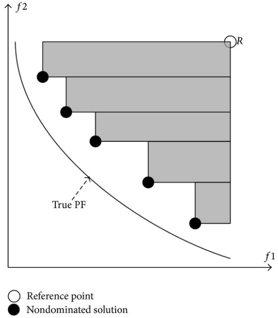
  Proof. "$\Rightarrow$" Suppose $\boldsymbol{x}^*$ is not an optimal solution of the ϵ-constraint problem. Then define $\epsilon = f(\boldsymbol{x}^*)$, and there exists $\boldsymbol{x} \in \mathcal{S}$ with $f_i(\boldsymbol{x}) < f_i(\boldsymbol{x}^*)$ and $f_j(\boldsymbol{x}) \leq \epsilon_j = f_j(\boldsymbol{x}^*)$, which contradicts the Pareto-optimality of $\boldsymbol{x}^*$.
  "$\Leftarrow$" Suppose $\boldsymbol{x}^*$ is not Pareto-optimal. Then there exists $\boldsymbol{x} \in \mathcal{S}$, such that $f_i(\boldsymbol{x}) < f_i(\boldsymbol{x}^*)$ for at least one $i \in \{1, \ldots, m\}$ and $f_j(\boldsymbol{x}) \leq f_j(\boldsymbol{x}^*)$ for all $j = 1, \ldots, m$. Since $\boldsymbol{x}^*$ is a solution of the ϵ-constraint problem, it follows that $f_j(\boldsymbol{x}) \leq f_j(\boldsymbol{x}^*) \leq \epsilon_j$, so $\boldsymbol{x}$ satisfies the constraints, which contradicts the optimality of $\boldsymbol{x}^*$ for the ϵ-constraint problem and index i.

EVALUATION OF SOLUTIONS

A common measure for the performance of a set of candidates $\mathcal{P} \subset \mathcal{S}$ is the **dominated hypervolume**:

$$S(\mathcal{P}, R) = \lambda\left(\bigcup_{\tilde{\boldsymbol{x}} \in \mathcal{P}} \{ \, \boldsymbol{x} \mid \tilde{\boldsymbol{x}} \prec \boldsymbol{x} \prec R \} \right),$$

where $\lambda(\cdot)$ is the Lebesgue measure and R a pre-specified reference point that every point in the set must dominate.



- HV is measured w.r.t. the reference point R, often chosen so each component is a known “worst” (upper bound) for that objective.
- HV is also called the **S-metric**.
- Its computation is $\mathcal{O}(n^{m-1})$ in the number of points and dimension m.
- Efficient algorithms exist for small m. In ML, we rarely exceed $m = 3$ in practice.